Chapter 8

DESCRIPTIVE STATISTICS AND PROBABILITY

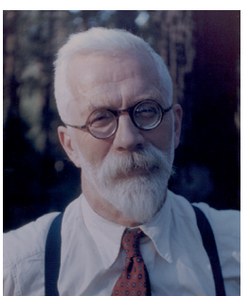
Learning Objectives

After studying this chapter students will be able to understand the following concepts.

- Measures of Central Tendency like A.M, G.M & H.M
- Relationship among the averages.
- Related positional measures like Quartiles, Deciles and percentiles etc.,
- Measures of Dispersion like Quartile Deviation, mean Deviation
- Relative measures like co-efficient of Quartile Deviation, co-efficient of Mean Deviation
- Concept of conditional probability and multiplication theorem.
- Baye's theorem and its applications.

8.1 Measures of Central Tendency

Introduction:


Sir Ronald Fisher

One of the most important objectives of Statistical analysis is to get one single value that describes the characteristic of the entire value for data. Such a value represent the measure of central tendency for the complete data. The word average is very commonly used in day-to-day conversation. For example, we often talk of average boy in a class, average height or average life of an Indian, average income, etc,. Sir Ronald Fisher who is known to be a father of statistics and he made his pioneering contributions in the applications of statistics in various disciplines.

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8.1.1 Average

- Recall

There are several measures of central tendency for the data. They are

- Arithmetic Mean
- Median
- Mode
- Geometric Mean
- Harmonic Mean

Arithmetic Mean (discrete case)

Arithmetic mean of a set of observations is their sum divided by the number of observations. The observation are classified into a) Ungrouped data and b) Grouped data.

a) Ungrouped data

(i) Direct Method:

$$\overline{X} = \frac{X_1 + X_2 + X_3 + \ldots X_n}{n} = \frac{\sum X}{n}$$

where $\overline{X}$ is Arithmetic Mean, $\sum X$ is sum of all the values of the variable X and n is number of observations.

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